

Lecture 20: Two-Stage Least Squares

Economics 326 — Introduction to Econometrics II

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Multiple linear IV model

- In practice, we often estimate models with multiple endogenous and exogenous regressors.
- Example:

$$\ln \text{Wage}_i = \gamma_0 + \gamma_1 \text{Age}_i + \gamma_2 \text{Sex}_i + \beta_1 \text{Educ}_i + \beta_2 \text{Children}_i + U_i.$$

- **Exogenous** regressors: age, sex, and a constant.
- **Endogenous** regressors: education and children (family size).

Multiple linear IV model

- General model:

$$y_i = \gamma_0 + \gamma_1 X_{1,i} + \dots + \gamma_k X_{k,i} + \beta_1 Y_{1,i} + \dots + \beta_m Y_{m,i} + U_i,$$

where

- y_i is the dependent variable.
- γ_0 is the intercept: $E[U_i] = 0$.
- $X_{1,i}, \dots, X_{k,i}$ are k **exogenous** regressors:

$$\text{Cov}(X_{1,i}, U_i) = \dots = \text{Cov}(X_{k,i}, U_i) = 0.$$

- $Y_{1,i}, \dots, Y_{m,i}$ are m **endogenous** regressors:

$$\text{Cov}(Y_{1,i}, U_i) \neq 0, \dots, \text{Cov}(Y_{m,i}, U_i) \neq 0.$$

Identification problem

- There are $k + 1 + m$ unknown coefficients:

$$y_i = \gamma_0 + \gamma_1 X_{1,i} + \dots + \gamma_k X_{k,i} + \beta_1 Y_{1,i} + \dots + \beta_m Y_{m,i} + U_i.$$

- The exogeneity conditions give only $k + 1$ equations:

$$E[U_i] = 0, \quad \text{Cov}(X_{j,i}, U_i) = 0, \quad j = 1, \dots, k.$$

- More unknowns than equations: the covariances between X 's, Y 's, and y do not suffice to recover the coefficients.
- Without additional information, the coefficients are **not identified** even at the population level.
- We need **at least** m additional equations!

Instrumental variables

- Suppose we observe l additional **exogenous** variables (IVs) $Z_{1,i}, \dots, Z_{l,i}$.
- The IVs are **excluded** from the **structural equation**:

$$y_i = \gamma_0 + \gamma_1 X_{1,i} + \dots + \gamma_k X_{k,i} + \beta_1 Y_{1,i} + \dots + \beta_m Y_{m,i} + U_i,$$

so there are still $k + 1 + m$ structural coefficients.

- Since the IVs are exogenous, we gain l additional moment conditions:

$$\text{Cov}(Z_{j,i}, U_i) = 0, \quad j = 1, \dots, l,$$

bringing the total to $k + 1 + l$ equations.

- **Necessary condition** for identification: $l \geq m$.

First-stage equations

- The IVs must also determine the endogenous regressors.
- The system consists of the **structural equation** and m **first-stage (reduced-form)** equations:

$$\begin{aligned} Y_{1,i} &= \pi_{0,1} + \pi_{1,1} Z_{1,i} + \dots + \pi_{l,1} Z_{l,i} \\ &\quad + \pi_{l+1,1} X_{1,i} + \dots + \pi_{l+k,1} X_{k,i} + V_{1,i}, \\ &\quad \vdots \\ Y_{m,i} &= \pi_{0,m} + \pi_{1,m} Z_{1,i} + \dots + \pi_{l,m} Z_{l,i} \\ &\quad + \pi_{l+1,m} X_{1,i} + \dots + \pi_{l+k,m} X_{k,i} + V_{m,i}. \end{aligned}$$

- The exogenous regressors X 's appear in the first-stage equations because they can be correlated with Y 's.
- The X 's and Z 's are **uncorrelated** with all errors U and V 's.

Order condition for identification

- Necessary condition: for every endogenous regressor Y , there must be **at least** one excluded exogenous variable Z :

$$l \geq m.$$

- $l = m$: the system is **exactly identified**.
- $l > m$: the system is **overidentified**.
- $l < m$: the system is **underidentified**; the structural coefficients γ 's and β 's cannot be estimated.

2SLS: the first stage

- The first-stage equations:

$$\begin{aligned} Y_{1,i} &= \pi_{0,1} + \pi_{1,1} Z_{1,i} + \dots + \pi_{l,1} Z_{l,i} \\ &\quad + \pi_{l+1,1} X_{1,i} + \dots + \pi_{l+k,1} X_{k,i} + V_{1,i}, \\ &\quad \vdots \\ Y_{m,i} &= \pi_{0,m} + \pi_{1,m} Z_{1,i} + \dots + \pi_{l,m} Z_{l,i} \\ &\quad + \pi_{l+1,m} X_{1,i} + \dots + \pi_{l+k,m} X_{k,i} + V_{m,i}. \end{aligned}$$

- All right-hand side variables are exogenous.
- The first-stage coefficients π 's can be estimated consistently by OLS, regressing each Y against Z 's and X 's.

2SLS: the first stage

- Let $\hat{\pi}$'s denote the OLS estimators of π 's. Obtain the fitted values:

$$\begin{aligned}\hat{Y}_{1,i} &= \hat{\pi}_{0,1} + \hat{\pi}_{1,1}Z_{1,i} + \dots + \hat{\pi}_{l,1}Z_{l,i} \\ &\quad + \hat{\pi}_{l+1,1}X_{1,i} + \dots + \hat{\pi}_{l+k,1}X_{k,i}, \\ &\vdots \\ \hat{Y}_{m,i} &= \hat{\pi}_{0,m} + \hat{\pi}_{1,m}Z_{1,i} + \dots + \hat{\pi}_{l,m}Z_{l,i} \\ &\quad + \hat{\pi}_{l+1,m}X_{1,i} + \dots + \hat{\pi}_{l+k,m}X_{k,i}.\end{aligned}$$

- The $\hat{Y}$'s are functions of Z 's and X 's (all exogenous), so they are asymptotically uncorrelated with the errors.

2SLS: the second stage

- In the second stage, regress (OLS) y against a constant, X 's, and the fitted values $\hat{Y}$'s:

$$\begin{aligned}y_i &= \hat{\gamma}_0^{2SLS} + \hat{\gamma}_1^{2SLS}X_{1,i} + \dots + \hat{\gamma}_k^{2SLS}X_{k,i} \\ &\quad + \hat{\beta}_1^{2SLS}\hat{Y}_{1,i} + \dots + \hat{\beta}_m^{2SLS}\hat{Y}_{m,i} + \hat{U}_i.\end{aligned}$$

- The 2SLS estimators $\hat{\gamma}_0^{2SLS}, \dots, \hat{\gamma}_k^{2SLS}, \hat{\beta}_1^{2SLS}, \dots, \hat{\beta}_m^{2SLS}$ are **consistent** and **asymptotically normal**.
- Standard errors from naïve second-stage OLS are **incorrect**: they do not account for the estimation error in $\hat{\pi}$'s from the first stage.
- Statistical packages report the corrected standard errors.

Example: returns to education

- Estimate the returns to education using the MROZ dataset (Wooldridge, 2006):

$$\ln \text{Wage}_i = \gamma_0 + \beta_1 \text{Educ}_i + \gamma_1 \text{Exper}_i + \gamma_2 \text{Exper}_i^2 + U_i.$$

- **Endogenous**: Educ (education).
- **Instruments**: MotherEduc and FatherEduc (parents' education).
- **Exogenous**: Exper and Exper^2 (experience).
- The system is **overidentified** ($l = 2 > m = 1$).
- In R, use `ivreg()` from the AER package with `vcovHC(..., type = "HC1")` for heteroskedasticity-robust standard errors.

Example: returns to education

First-stage regression (Educ on instruments and exogenous regressors):

```
first_stage <- lm(educ ~ exper + expersq + motheduc + fatheduc, data = d)
coeftest(first_stage, vcov = vcovHC(first_stage, type = "HC1"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.1026401	0.4241444	21.4612	< 0.00000000000000022 ***
exper	0.0452254	0.0419107	1.0791	0.2812
expersq	-0.0010091	0.0013233	-0.7626	0.4461
motheduc	0.1575970	0.0354502	4.4456	0.00001121343 ***
fatheduc	0.1895484	0.0324419	5.8427	0.00000001026 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Both instruments (motheduc, fatheduc) are statistically significant. First-stage $R^2 = 0.2115$.

Example: returns to education

2SLS second stage:

```
iv_fit <- ivreg(lwage ~ educ + exper + expersq |  
               exper + expersq + motheduc + fatheduc, data = d)  
coeftest(iv_fit, vcov = vcovHC(iv_fit, type = "HC1"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.04810031	0.42979771	0.1119	0.910945
educ	0.06139663	0.03333859	1.8416	0.066231 .
exper	0.04417039	0.01554638	2.8412	0.004711 **
expersq	-0.00089897	0.00043008	-2.0902	0.037193 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

OLS for comparison:

```
ols_fit <- lm(lwage ~ educ + exper + expersq, data = d)  
coeftest(ols_fit, vcov = vcovHC(ols_fit, type = "HC1"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.52204056	0.20165046	-2.5888	0.009961 **
educ	0.10748964	0.01321897	8.1315	0.00000000000000472 ***
exper	0.04156651	0.01527304	2.7216	0.006765 **
expersq	-0.00081119	0.00042007	-1.9311	0.054139 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The 2SLS estimate of the return to education ($\hat{\beta}_1^{2SLS} = 0.061$) is smaller than the OLS estimate ($\hat{\beta}_1^{OLS} = 0.107$), consistent with upward ability bias.